

DOMAS Calibrated Scores — Model Cards

DOMAS calibrated scores — model cards

DOMAS emits two **calibrated probability** scores alongside the categorical **impact** level, one per axis of alternative-splicing consequence:

score	axis	question it answers	gene-grouped AUC
fold_change_prob	structural	Does the alternative isoform adopt a <i>different 3-D fold</i> ?	0.894
impact_prob	functional	Is the changed region <i>functionally important</i> (pathogenic-variant-like)?	0.766

They are deliberately different questions ("fold-vs-function duality"): a change can preserve the fold yet abolish function, or wreck the fold while being biologically silent. Each score is a logistic regression over a small set of interpretable features, **natively calibrated** (a predicted 0.6 really means ~60%), reported throughout under **gene-grouped 5-fold cross-validation** (every isoform of a gene is confined to one fold, so no protein appears in both train and test — no leakage).

Everything below is reproduced end-to-end by `reproduce_models.py` (see §5).

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1. Shared benchmark, labels, and methodology

Benchmark dataset. Song et al., *Predicting the structural impact of human alternative splicing*, **Genome Biology** 2025 (doi:10.1186/s13059-025-03744-x). One row = **one canonical-vs-alternative-isoform splicing event**. After dropping pure insertions (no canonical changed region to score), **10,227 pairs across 5,557 human proteins**.

Label sources — what they are and what they mean:

label	source	meaning	used by
TM-score	The paper folded every canonical and alternative isoform with AlphaFold2 and computed the pairwise TM-score (Table S4).	Structural-similarity of two folds, 0–1. TM < 0.5 = different fold ; ≥ 0.5 = same fold . The structural ground truth.	fold_change_prob
humsavar	UniProt humsavar disease-variant catalogue (aggregates ClinVar/OMIM), single-residue variants in canonical protein coordinates , classed LP/P (pathogenic) or LB/B (benign).	A changed region "overlaps a pathogenic/benign variant" if a LP/P / LB/B position falls inside it.	impact_prob

Evaluation protocol. L2-regularised logistic regression; features standardised (z-scored); missing features imputed to the training median. Metrics:

- **AUC** — ROC-AUC (rank/threshold-independent; 0.5 = chance).
- **accuracy** — at decision threshold 0.5.
- **R²** — for fold_change_prob, variance of the *continuous* TM-score explained by a ridge regressor on the same features; for impact_prob (a binary target), the variance of the 0/1 label explained by the predicted probability (label-R², lower by construction).

Feature provenance (each feature, and where its value comes from):

feature	what it is	source
pae_global	mean Predicted Aligned Error over the whole <i>canonical</i> AlphaFold structure — how rigidly/confidently the fold is assembled	AlphaFold DB predicted_aligned_error_v6.json → afdp_pae
identity	% sequence identity between canonical and alternative protein, from the trimmed changed region	UniProt reference proteome + varsplc sequences
max_cov_loss	maximum Pfam domain coverage (%) lost in the changed region	HMMER hmmsearch --cut_ga vs Pfam-A → pfam_match
protL	canonical protein length (aa)	UniProt sequence
region_am	mean AlphaMissense pathogenicity over the changed region	AlphaMissense per-residue → am_pathogenicity
loeuif	gnomAD LOEUF gene loss-intolerance (lower = more constrained)	gnomAD v2.1.1 by-gene → gene_constraint

feature	what it is	source
buried_fraction	fraction of the changed region buried (RSA < 0.25) in the canonical structure	AlphaFold DB model → Shrake–Rupley RSA → afd_buried

1.1 Two training / evaluation methods (and their agreement)

The script evaluates each model two ways so the honesty of the headline numbers can be checked:

- **Method 1 — gene-grouped 5-fold CV** (the reported/shipped method). All isoforms of a gene are confined to a single fold, so no protein appears in both train and test. This is the leakage-free estimate and matters because a gene-level feature (l_oef) and correlated same-gene isoforms could otherwise leak.
- **Method 2 — random 80/20 hold-out** (pair-level). A single random 80% train / 20% test split, ignoring gene membership. This is the naive method; if it scored much higher than Method 1 it would reveal same-gene leakage.

They agree closely — the small gap confirms leakage is minimal:

model	metric	Method 1 (gene-grouped CV)	Method 2 (random 80/20)	Δ (holdout – grouped)
fold_change_prob	AUC	0.894	0.898	+0.005
	accuracy	0.815	0.816	+0.001
	R ²	0.659	0.677	+0.018
impact_prob	AUC	0.766	0.778	+0.012
	accuracy	0.750	0.730	–0.020
	R ²	0.175	0.189	+0.014

The random split is *slightly* optimistic on AUC/R² ($\leq +0.018$), the expected mild effect of same-gene pairs landing in both train and test; accuracy even drops for `impact_prob` (single-split noise on 655 test rows). All gaps are ≤ 0.02 , so the two methods give essentially the same answer — **the gene-grouped numbers are honest, not inflated by leakage**. They are reported as the headline because they are the conservative estimate; Method 2 is a single-split point estimate (higher variance) shown only for contrast.

2. fold_change_prob — structural axis

What it measures. Calibrated P(the alternative isoform adopts a **different fold**, TM-score < 0.5).

How it's measured / label. Trained and evaluated against the paper's AlphaFold2 TM-score (§1). TM < 0.5 is the positive class.

Features & standardized coefficients (utils._FOLD_CHANGE_MODEL; larger |coef| = more influence):

feature	std. coef	direction
pae_global	+1.96	floppier / multi-domain canonical structure → more fold change (dominant)
identity	−0.68	more sequence conserved → fold preserved
max_cov_loss	+0.65	more Pfam domain lost → more fold change
protL	−0.54	longer protein → fold preserved (larger scaffold)
intercept	−1.26	

(raw feature medians [17.13, 75.0, 11.0, 388.0], means [16.59, 66.36, 33.22, 379.18], std [7.12, 28.02, 39.28, 134.36] for the four features in order.)

Performance. Base rate $P(\text{TM} < 0.5) = 0.338$, so the majority-class accuracy baseline is 0.662.

operating point	AUC	accuracy	R ²	coverage
all events (no band)	0.894	0.815	0.659	100%
suggested band applied (route 0.40–0.60 to folding)	0.913	0.854	0.686	88% called

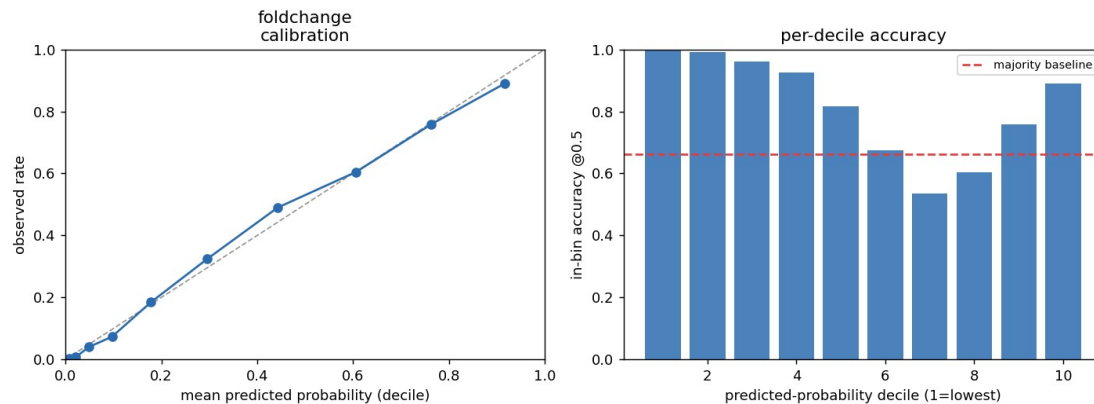
Suggested query band: route fold_change_prob ∈ [0.40, 0.60] to real folding

(ESMFold/ColabFold) instead of forcing a call. Rationale, visible in the decile table and the right-hand graph below: the middle deciles (7–8, prob ≈ 0.37–0.69) are where the model degrades to a **coin flip** — in-bin accuracy dips to 0.54, *below* the 0.662 majority baseline — because these are the "does the remainder re-fold?" cases (the SRP9 class) that cheap features provably cannot resolve. Abstaining on that ~12% lifts the called-set accuracy 0.815 → 0.854 and AUC 0.894 → 0.913. Widen to 0.30–0.70 for higher-stakes screening.

Decile calibration (events sorted by predicted probability into 10 equal bins):

decile	prob range	mean pred	observed rate	n	in-bin acc@0.5
1	0.001–0.013	0.007	0.004	1023	0.996
2	0.013–0.032	0.022	0.008	1023	0.992
3	0.032–0.067	0.048	0.040	1023	0.960
4	0.067–0.133	0.097	0.073	1023	0.927
5	0.133–0.230	0.178	0.185	1023	0.815
6	0.230–0.367	0.296	0.325	1023	0.675
7	0.367–0.524	0.443	0.490	1023	0.536
8	0.524–0.685	0.606	0.605	1022	0.605
9	0.685–0.847	0.763	0.758	1022	0.758
10	0.848–0.991	0.917	0.890	1022	0.890

Predicted $\approx$ observed in every decile (well-calibrated). The U-shaped accuracy — high at both confident tails, coin-flip in the middle — is the signature that motivates the band.



Caveats.

- **Confounded mechanism.** Part of `pae_global`'s power is that a TM-score between two *low-confidence* AlphaFold models is low almost by construction. So a high canonical PAE partly flags "this TM is unreliable" as much as "a real fold change occurred." It is still non-circular (canonical-only, computed independently of the label) and prospectively available, but read it as "confidence the fold is well-defined enough to compare," not a pure fold-change detector.
- **Per-protein prior.** `pae_global` is a whole-canonical-structure property, identical for every isoform of a gene — it sets a per-protein baseline; only `identity`, `max_cov_loss`, `protL` differentiate isoforms of the same gene.
- **Known blind spot.** High-identity / low-TM outliers ($\approx 10\%$ of pairs: identity $\geq 80\%$ yet TM < 0.5 — the SRP9 class) are confidently mis-called *preserved*; these sit *outside* the routing band, so pair the band with a "high-identity AND exposed region" trigger for high-stakes use.
- **Scope.** Pure insertions are excluded (no canonical changed region). The ceiling for cheap features is $\sim$ AUC 0.90; the residual genuinely needs the isoform folded.

3. `impact_prob` — functional axis

What it measures. Calibrated $P(\text{the changed region is pathogenic-variant-like})$. Concretely, among changed regions that overlap *some* curated variant, $P(\text{it is the pathogenic one rather than the benign one})$. It is an **enrichment / prioritisation** signal ("look here first"), **not** a per-event clinical classifier.

How it's measured / label. Restricted to regions overlapping a humsavar LP/P or LB/B variant; label = 1 if pathogenic-overlapping, 0 if only benign-overlapping. 3,276 regions. (The benign class is the specificity control — a real functional score must enrich pathogenic, not merely variant-dense, regions.)

Features & standardized coefficients (`utils` impact model):

feature	std. coef	direction
region_am	+0.80	higher AlphaMissense constraint → pathogenic (dominant)
loeuf	-0.42	more loss-intolerant gene (lower LOEUF) → pathogenic
buried_fra c	+0.24	buried / structured region → pathogenic
max_cov_lo ss	-0.06	~no effect
intercept	-1.19	

(raw medians [0.482, 0.983, 41.0, 0.340], means [0.482, 0.979, 46.07, 0.309], std [0.155, 0.443, 41.01, 0.219].)

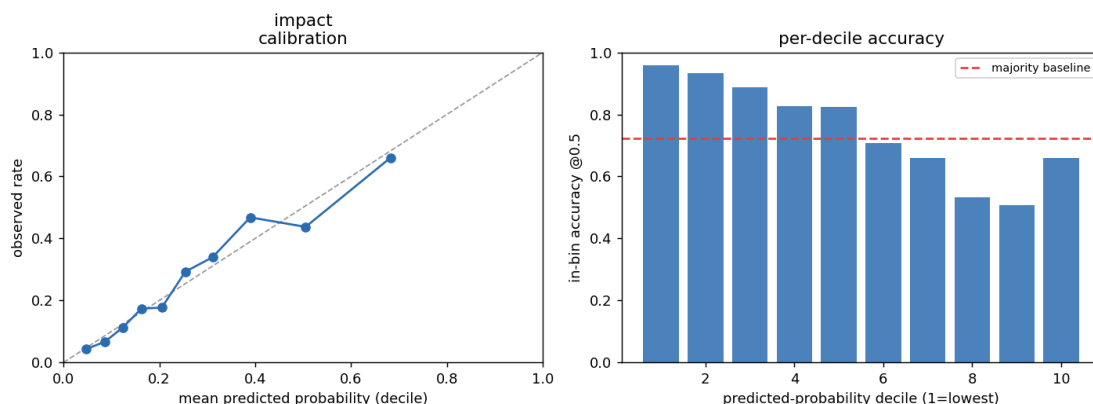
Performance. Base rate $P(\text{pathogenic} \mid \text{overlaps a variant}) = 0.277 \rightarrow$ majority baseline 0.723.

operating point	AUC	accuracy	R ²	coverage
all events (no band)	0.766	0.750	0.175	100%
abstain on 0.40–0.60	0.770	0.793	0.181	85% called

Suggested use / band. Treat `impact_prob` as a **triage/prioritisation** score, not a verdict: the **top decile (prob ≥ 0.58)** is ~66% pathogenic — "review these first"; the **bottom deciles (prob ≤ 0.11)** are ~5–7% pathogenic — reliably de-prioritised; the **0.40–0.60** middle is genuinely ambiguous and should go to manual curation. Abstaining on the middle raises called-set accuracy 0.750 → 0.793 (AUC barely moves — the tails are where the signal is). Do **not** use it to call an individual variant pathogenic.

Decile calibration:

decile	prob range	mean pred	observed rate	n	in-bin acc@0.5
1	0.019–0.068	0.047	0.043	328	0.957
2	0.068–0.107	0.087	0.067	328	0.933
3	0.107–0.143	0.124	0.113	328	0.887
4	0.143–0.184	0.163	0.174	328	0.826
5	0.184–0.227	0.205	0.177	328	0.823
6	0.227–0.281	0.254	0.293	328	0.707
7	0.281–0.345	0.311	0.339	327	0.661
8	0.345–0.436	0.390	0.468	327	0.532
9	0.438–0.576	0.505	0.437	327	0.508
10	0.576–0.911	0.683	0.661	327	0.661



Caveats.

- **Semi-circular (the biggest one).** `region_am` (AlphaMissense) is itself a pathogenicity predictor and dominates the model (+0.80), so part of the signal is built in. The **non-circular** contributors are `loeufr` (gnomAD constraint) and `buried_frac` (structure). Read the score as "concentrates known functional signals," not independent evidence.
- **Positive-only labels.** "No known pathogenic variant" is not proof of neutrality — so these are recall/enrichment results, not true specificity.
- **Region-length confound.** On *raw* variant-overlap presence, region length dominates (bigger change → more residues → more overlap); `impact_prob` is trained length-controlled (patho-vs-benign among overlapping regions) precisely to avoid measuring size instead of function. Do not add sequence-identity/length as a feature — it would inflate AUC while measuring region size.
- **Not a classifier.** Both classes pile into moderate/high probabilities; specificity is limited. It ranks/prioritises; it does not adjudicate.

4. Why these features and not others (audit trail)

Both feature sets are the survivors of an exhaustive search (EXPERIMENTS.md, E1–E44). Key negatives worth knowing for review:

- **PAE dominates fold change** and was the single biggest lever (AUC 0.78 → 0.90); it was *assumed* redundant for years and only helped when finally measured.
- **buried_frac is functional-only; pae_global is structural-only** — each was tested on the *other* axis and rejected (drop-column ≈ 0 or fails the benign control). The two models are cleanly separated.
- **Rejected / redundant:** ESM-2 embeddings (help fold change but not shipped — heavy), 650M vs 150M ESM (≈equal), region pLDDT (0.79-correlated with burial), contacts (redundant with burial), phyloP (only a small structural add, small sample), peak-AM, tree/NN model families, multi-task. None beat the shipped logistic on its own axis.

5. Reproduction

`reproduce_models.py` reproduces every number and both graphs in this document. It:

1. **Downloads** (into `./repro_data/`, ~30 MB, no DoChAP DB needed): UniProt human reference proteome + varsplic (sequences → identity, protL, changed region), UniProt humsavar (functional labels), gnomAD v2.1.1 by-gene constraint (loeuf).
2. **Reuses** the committed per-pair feature CSVs in this folder (their raw provenance — AlphaFold DB, AlphaMissense, Pfam — is documented in `DATA_SOURCES.md`): `full_pairs.csv` (TM label + changed region), `pae_features_full.csv` (`pae_global`), `bench_rich_results.csv` (`region_am`, `max_cov_loss`), `full_features.csv` (`buried_frac`).
3. **Trains + tests** both models by both methods of §1.1 (gene-grouped CV and random 80/20), prints coefficients (which match the shipped `utils` models), overall + banded metrics, the method comparison, and the decile tables, and writes `model_card_foldchange.png` / `model_card_impact.png`.

```
cd alphafold_benchmark && python3 reproduce_models.py          #
everything, both models, both methods
python3 reproduce_models.py --download-only                    #
just fetch inputs into repro_data/ and exit
python3 reproduce_models.py --models fold_change --methods grouped  #
one model, one method
python3 reproduce_models.py --methods holdout --seed 7          #
only the random 80/20 split, seed 7
python3 reproduce_models.py --no-graphs                         #
skip the PNGs
```

Flags: `--models {fold_change,impact,both}`, `--methods {grouped,holdout,both}`, `--download-only`, `--no-graphs`, `--seed N` (hold-out split seed).

6. Review checklist / essentials

- **Intended use.** Screening/prioritising alternative-splicing events at scale: `fold_change_prob` for "does the fold change," `impact_prob` for "is the changed region functionally important." Both are triage signals feeding human/structural review.
- **Out of scope.** Not a clinical variant classifier; not validated on non-human proteins; not for pure insertions; not a substitute for folding the isoform when the score is in the uncertain band.
- **Leakage control.** Gene-grouped CV throughout (all isoforms of a gene in one fold); loeuf is gene-level so pair-level CV would leak — the gene-grouped numbers are the honest ones.
- **Class balance.** fold change: 33.8% positive (majority acc 0.662). functional: 27.7% pathogenic among overlapping regions (majority acc 0.723). Always compare model accuracy to these baselines, not to 0.5.

- **Calibration.** Both are logistic (natively calibrated); the decile tables show predicted $\approx$ observed in every bin.
 - **Known failure modes.** fold change: high-identity/low-TM re-folders (SRP9 class), outside the band. functional: both variant classes pile high (limited specificity); semi-circular via AlphaMissense.
 - **Versioning.** Coefficients live in `code/utils.py` (`_FOLD_CHANGE_MODEL` and the impact model), regenerated by `fit_foldchange_pae.py` / `fit_calibrated.py`. Card reflects the current shipped models; base benchmark = Genome Biology 2025.
 - **Provisioning dependency.** `fold_change_prob` needs the `afdb_pae` table (`build.py -source afdb_pae`); if absent, `pae_global` imputes to the training median and the score degrades gracefully toward an identity/coverage-only prediction.
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Appendix A — Every feature evaluated, and where it comes from

The two shipped feature sets are the survivors of the full investigation (EXPERIMENTS.md, E1–E44). This appendix catalogues **all** features that were built and tested — shipped and rejected — grouped by data source, with what each is and why it did or didn't make the cut. "S" = in `fold_change_prob` (structural); "F" = in `impact_prob` (functional).

Sequence-derived (UniProt reference proteome + `uniprot_sprot_varsplc`):

feature	what it is	status
<code>identity</code>	% sequence identity between canonical and alternative protein, from the trimmed changed region	shipped (S) — #2 structural feature
<code>protL</code>	canonical protein length (aa)	shipped (S) — small real add
<code>reglen</code>	length (aa) of the changed region	dropped — it is the region-length confound; deliberately excluded (would measure size, not function)

Canonical-AlphaFold-structure-derived (AlphaFold DB model + its confidence files):

feature	what it is	source	status
<code>pae_global</code>	mean Predicted Aligned Error over the whole canonical structure (fold rigidity/confidence)	AFDB <code>predicted_aligned_error_v6.json</code> → <code>afdb_pae</code>	shipped (S) — dominant feature (the biggest lever, AUC 0.78 → 0.90)
<code>buried_frac</code>	fraction of the changed region buried (RSA < 0.25)	AFDB model → Shrake–Rupley RSA → <code>afdb_rsa</code>	shipped (F) — functional (buried → pathogenic-rich)

feature	what it is	source	status
mean_rsa	mean relative solvent accessibility of the changed region	same as above	dropped — near-duplicate of buried_frac
pae_reg2rest	mean PAE between the changed region and the rest of the fold (region-specific rigidity)	AFDB PAE	dropped — strong alone (AUC 0.73) but redundant with pae_global in the full model
region_plddt	mean per-residue pLDDT of the changed region (local order/confidence)	AFDB model b-factors → afdb_plddt	dropped — 0.79-correlated with buried_frac; +0.001 functional / +0.002 structural
helix_frac / strand_frac / structured_frac	DSSP secondary-structure fractions of the changed region	AFDB model → pydssp	dropped — small, subsumed by burial/PAE
contact density	C β <8 Å contacts from the changed region to the rest of the fold	AFDB model coordinates	dropped — redundant with burial (partial corr collapses after burial)
dist_term / rel_center / frac_before / is_terminal	position of the changed region relative to the protein termini	changed-region coords + protL	dropped — +0.017 structural <i>before</i> PAE, but PAE absorbs it (+0.002 after)

Isoform-AlphaFold-structure-derived (from Song et al. Genome Biology 2025 Table S4 — computed from the *isoform's* own AF2 fold, hence **circular** for predicting TM):

feature	what it is	status
iso helix/sheet/loop, radius of gyration, surface charge, IDR %	global structural descriptors of the alternative isoform's predicted fold	dropped — predict TM at AUC 0.92 but circular (need the isoform folded, i.e. already have TM)
PTM-change counts (ptm_added, buried ↔ exposed, ptm_missed)	how the isoform gains/loses/relocates canonical PTM sites	dropped — non-circular but +0.002 functional (redundant with region_am/length)
charge / radius deltas	canonical-vs-isoform change in charge / radius of gyration	dropped — ~0 contribution

Protein language model (ESM-2, run over the sequences):

feature	what it is	status
ESM-2 150M / 650M region + (canonical-alt) difference embeddings (mean-pooled, PCA-50)	learned sequence representation of the changed region	tested — <i>breaks the structural ceiling</i> (AUC 0.79 → 0.82) but heavy to provision; not shipped . 650M ≈ 150M.
ESM multi-pool (mean+max+std+first+last token)	positional pooling of per-residue embeddings	dropped — equal to mean-pooling

Variant / constraint databases:

feature	what it is	source	status
region_am	mean AlphaMissense pathogenicity over the changed region	AlphaMissense → am_pathogenicity	shipped (F) — dominant functional feature
loeuif	gnomAD gene loss-intolerance (lower = more constrained)	gnomAD v2.1.1 by-gene → gene_constraint	shipped (F) — non-circular constraint signal
pLI	gnomAD probability of LoF-intolerance	gnomAD	dropped — weaker than LOEUF
peak-AM / mean-of-top-K AM	hotspot AlphaMissense instead of regional mean	AlphaMissense	dropped — regional mean beats every peak (domain-scale regions)
func_site	whether the region overlaps a curated UniProt functional residue	UniProt features	used in the <i>categorical impact</i> only (semi-circular for the probability)

Domain / coverage:

feature	what it is	source	status
max_cov_loss	maximum Pfam domain coverage (%) lost in the changed region	HMMER hmmsearch --cut_ga vs Pfam-A → pfam_match	shipped (S and F) — the one feature in both models (opposite signs)
fold_state	folded / disordered, from UniProt HELIX/STRAND/disorder annotation	UniProt features	used upstream in the categorical impact

Cross-species conservation:

feature	what it is	source	status
phyloP (region / whole-protein mean)	evolutionary conservation of the changed region	UCSC hg38 100-way phyloP bigWig + EBI coordinates API	tested — a small structural add (+0.037 AUC on a 363- pair sample); not shipped pending scale-up

Not available: NMD (nonsense-mediated decay) — DoChAP filters NMD-targeted transcripts at DB-build time, so degraded isoforms never reach the benchmark; NMD is neither a feature nor a confound here.

Appendix B — Comparable published methods

We looked for published methods attempting each task and compare them below. A recurring theme: for the **structural** axis the prior work *folds every isoform* (expensive) and *catalogues* the result, whereas DOMAS *predicts* the fold-change label from cheap canonical-only features; for the **functional** axis the closest analogues score exon / isoform importance from protein + conservation + expression features, on labels that are related but not identical to ours (so AUCs are not head-to-head).

B.1 Structural axis — fold_change_prob

method	approach	reported result	relation to ours
Song et al., <i>Genome Biology</i> 2025 (our label source)	Fold all 11,161 human isoforms with AlphaFold2 ; compare each to its canonical with TM-score + 5 other metrics (secondary structure, surface charge, radius of gyration, PTM SASA, IDR).	Structural similarity largely tracks sequence identity; flags 328 high-identity / low-TM outliers (53 with clear domain alterations). A <i>characterisation</i> , not a predictor.	This is exactly the TM ground truth we predict. We reproduce their label from 4 cheap canonical features (no isoform folding) at AUC 0.90 / R² 0.66 , and quantify the high-id/low-TM blind spot they identified.
Yang et al., <i>bioRxiv</i>	AF2 structures of AS	AS regions are	Supports our finding

method	approach	reported result	relation to ours
2024.01.30.578053	isoforms; statistical characterisation of where AS falls.	enriched in loops / exposed / disordered residues ; qualitative case studies (Septin-9, Tau). No TM predictor, no isoform-vs-isoform structural score.	that exposed/peripheral changes drive low TM; provides no quantitative predictor to compare against.
ASpdb (Nucleic Acids Research, Database issue 2025)	Knowledgebase of >3,400 canonical + >7,200 alternative-isoform AF2 models with comparative structural-alteration calls and visualisation.	A resource, not a predictive model; provides a second, non-TM structural-alteration label.	Complementary independent label source (flagged in our EXPERIMENTS E40 as a future cross-check; its calls are per-entry web content, not bulk-downloadable).
DIGGER / DIGGER 2.0 (Louadi et al., NAR 2021 / 2025)	Augment the protein–protein interaction network with domain–domain interactions + splicing; identify which interactions/domains an AS event disrupts (interaction rewiring).	Network/rule-based (no AUC); maps exon → lost domain → lost interactions.	Different structural readout — <i>interaction</i> consequence, not global fold. Closest in spirit to our max_cov_loss (domain loss) feature; complementary to a fold-change probability.

Bottom line (structural). No prior method offers a cheap *predictor* of the fold-change (TM) label — they fold the isoform or catalogue outcomes. DOMAS's contribution is a calibrated ~AUC-0.90 predictor from canonical-only features, with **pae_global (PAE)** as the dominant signal — to our knowledge not previously used for this task.

B.2 Functional axis — *impact_prob*

method	approach	reported result	relation to ours
ExonImpact (2017, PMC5390777)	Random forest on protein-level features of the spliced exon: predicted disorder (SPINE-D), ASA + secondary structure (SPINE-X), Pfam-domain overlap,	10-fold CV AUC 0.83 –0.835 . Positives = 4,211 HGMD splice-disrupting exons; negatives = 2,664 1000-Genomes in-frame indel exons.	The closest analogue. Higher AUC, but on an easier, different label (disease-exon vs benign-indel-exon, with exon length as a feature) and using <i>predicted</i> disorder/ASA where we use

method	approach	reported result	relation to ours
	PTM sites (dbPTM), conservation, exon length.		AlphaFold-derived burial. Our label (pathogenic-vs-benign among variant-overlapping regions, length-controlled, gene-grouped) is harder — AUCs are not head-to-head.
TRIFID (Pozo et al., <i>NAR Genomics & Bioinformatics</i> 2021)	Gradient-boosted trees scoring the functional relevance of each splice isoform (0–1) from cross-species conservation, expression, and APPRIS/Pfam-integrity features.	Conservation is the top predictor; high-scoring isoforms' exons are under purifying selection, low-scoring ones neutral. Bimodal scores. Integrated into APPRIS.	Isoform-level (not region-level) and no single AUC against a variant label. Its top signal (conservation) is the axis we tested as phyloP (small add). Complementary; different granularity.
pext (Cummings et al., <i>Nature</i> 2020)	Not ML: GTEx transcript-expression -based per-base annotation (proportion expressed across transcripts) — is the region actually expressed?	Low-pext filtering removes 22.8% of pLoF in haploinsufficient genes but only 3.8% of ClinVar pathogenic variants.	Orthogonal axis (expression) to ours (constraint + structure). We <i>considered but did not use</i> GTEx (E41); pext is the strong published version of that idea and is complementary to impact_prob .

Not direct comparators. SpliceAI / MMSplice / AbSplice predict *whether a variant alters splicing* (the upstream event), not the functional consequence of the resulting spliced region — a different task one level up. AlphaMissense scores single missense variants; we *consume* it as **region_am** and operate at the region level.

Bottom line (functional). DOMAS's **impact_prob** sits in the same family as ExonImpact and TRIFID (protein/constraint features → functional-importance score) but is **region-level, calibrated, gene-grouped, and length-controlled**, and it is explicitly paired with the orthogonal structural score. Its AUC (0.766) is lower than ExonImpact's headline 0.83, but on a deliberately harder, confound-controlled label; the honest framing throughout is that it is an **enrichment/prioritisation** signal, not a classifier.

Sources. Song et al., *Genome Biology* 2025 · Yang et al., *bioRxiv* 578053 · ASpdb, *NAR* 2025 · DIGGER 2.0, *NAR* 2025 · ExonImpact, PMC5390777 · TRIFID, *NAR Genom. Bioinform.* 2021 · pext / Cummings et al., *Nature* 2020